CORTEZO, MARKY G. **MARCH 26, 2025**

BSIT 2-1 **COMP 009 – Object Oriented Programming**

Midterm Activity #3 – Simple Wallet System

Program Requirements

1. Create a Java program named: SimpleWalletLastName

(Example: SimpleWalletBedis)

2. Assume the initial balance is: 3,000 pesos

3. Display the following menu:

===== WALLET MENU =====

1. View Balance

2. Add Money

3. Spend Money

4. Exit

Additional Requirement

✓ Count and display the number of successful transactions only

Program Rules

Your program must use ONLY:

• ✓ Scanner class (for input)

• ✓ while loop (continuous execution)

• ✓ switch-case (menu selection)

• ✓ if-else (validation)

• ✓ counter variable

Important Conditions

Spend Money Rules:

• Maximum spend per transaction: 1,000 pesos

• Do NOT allow spending if:

o Amount exceeds balance

o Amount exceeds 1,000

SOURCE CODE:

```
import java.util.Scanner;

public class SimpleWalletCortezo {

    public static void main(String[] args) {

        // Setup scanner to read user input
        Scanner scanner = new Scanner(System.in);

        // Starting balance
        int balance = 3000;

        // This counts how many successful transactions happened
        int count = 0;

        // This controls the while loop
        // If true = keep looping, if false = stop
        boolean running = true;

        // Keep showing the menu until the user picks Exit
        while (running) {

            // Show the menu
            System.out.println("\n===WALLET MENU===");
            System.out.println("1. Check Balance");
            System.out.println("2. Add Money");
            System.out.println("3. Spend Money");
            System.out.println("4. Exit");
            System.out.print("Choice: ");

            // Read what the user picked
            int choice = scanner.nextInt();

            // Check what the user picked using switch-case
            switch (choice) {

                case 1:
                    // User wants to check balance
                    System.out.println("Output: Balance = " + balance);
                    count = count + 1; // add 1 to the counter
                    break;

                case 2:
                    // User wants to add money
                    System.out.print("Add: ");
                    int addAmount = scanner.nextInt();

                    // Check if the amount is valid
                    if (addAmount > 0) {
                        // Valid amount, so add it to balance
                        balance = balance + addAmount;
                        System.out.println("Output: New Balance = " + balance);
                        count = count + 1; // add 1 to the counter
                    } else {
                        // Invalid amount
                        System.out.println("Output: Invalid amount. Must be greater than 0.");
                    }
                    break;
```

```
                case 3:
                    // User wants to spend money
                    System.out.print("Spend: ");
                    int spendAmount = scanner.nextInt();

                    // Rule 1: Amount must be greater than 0
                    if (spendAmount <= 0) {
                        System.out.println("Output: Invalid amount. Must be greater than 0.");

                    // Rule 2: Cannot spend more than 1000 pesos at a time
                    } else if (spendAmount > 1000) {
                        System.out.println("Output: Exceeds maximum spend");

                    // Rule 3: Cannot spend more than current balance
                    } else if (spendAmount > balance) {
                        System.out.println("Output: Insufficient balance");

                    } else {
                        // All rules passed, spending is allowed
                        balance = balance - spendAmount;
                        System.out.println("Output: New Balance = " + balance);
                        count = count + 1; // add 1 to the counter
                    }
                    break;

                case 4:
                    // User wants to exit
                    System.out.println("Transaction Count: " + count);
                    System.out.println("Exiting program...");
                    running = false; // this stops the while loop
                    break;

                default:
                    // User typed something that is not 1, 2, 3, or 4
                    System.out.println("Output: Invalid choice. Please select 1-4.");
            }

        } // end of while loop

        // Close the scanner when done
        scanner.close();

    } // end of main method

} // end of class
```

Sample Execution:

```
<terminated> SimpleWalletCortezo [Java Application] C:\Users\Marky\.p2\pool\plugins\org.eclipse.justj.openjdk.h

===WALLET MENU===
1. Check Balance
2. Add Money
3. Spend Money
4. Exit
Choice: 1
Output: Balance = 3000

===WALLET MENU===
1. Check Balance
2. Add Money
3. Spend Money
4. Exit
Choice: 2
Add: 2000
Output: New Balance = 5000

===WALLET MENU===
1. Check Balance
2. Add Money
3. Spend Money
4. Exit
Choice: 3
Spend: 1500
Output: Exceeds maximum spend. Must below 1000

===WALLET MENU===
1. Check Balance
2. Add Money
3. Spend Money
4. Exit
Choice: 3
Spend: 500
Output: New Balance = 4500
```

```
===WALLET MENU===
1. Check Balance
2. Add Money
3. Spend Money
4. Exit
Choice: 5
Output: Invalid choice. Please select 1-4.

===WALLET MENU===
1. Check Balance
2. Add Money
3. Spend Money
4. Exit
Choice: 4
Transaction Count: 4
Exiting program...
```